

Don’t Be Choosy: Scoring over Choosing for Verbalized LLM Confidence

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Abstract

Accurate uncertainty estimation is critical for the reliable deployment of large language models (LLMs). A simple and widely used approach is to ask a model to verbalize its confidence on each item, referred to as **AbsScore** henceforth. However, AbsScores can be poorly discriminative, exhibiting overconfidence or collapsing to a small set of preferred values. Recent work proposed pairwise elicitation as an alternative, asking the model which of two items it is more confident about. While effective, this formulation forces a preference even when confidences are similar and discards information about the magnitude of the confidence difference. To overcome these limitations, we introduce **PairScore**, a simple alternative: present two items together, but ask the model to score both rather than choose between them. This preserves graded confidence while retaining the benefits of pairwise elicitation, and the resulting scores can be aggregated directly without fitting a separate ranking model. Across diverse models and tasks, including claim verification and best-of- N response selection, PairScore provides a strong alternative to both AbsScore and PairChoice. We further find that PairScore can distinguish items whose AbsScores collapse, while remaining robust to the choice of paired item. Together, these results demonstrate that PairScore offers a simple and effective way to elicit more informative confidence from LLMs.

1 Introduction

Large language models (LLMs) are increasingly deployed in high-stakes settings such as health, law, and decision support, where errors can carry substantial costs. A practical safeguard is to elicit a *confidence* or *uncertainty* estimate alongside each

model output, allowing low-confidence predictions to be flagged, filtered, or deferred to human review (Chen et al., 2023; Gupta et al., 2024). A simple, low-cost, and widely used approach is direct verbalization (**AbsScore**): prompting the model to express its confidence as a numerical score, for example on a 0–100 scale (Kadavath et al., 2022; Tian et al., 2023; Yang et al., 2024).

However, verbalized numerical confidence has well-documented limitations: scores can cluster near scale endpoints, shift with prompt phrasing, and vary across repeated queries (Shrivastava et al., 2023; Tian et al., 2023; Xiong et al., 2024). Such pathologies can make the resulting confidence scores less informative for distinguishing and ranking model outputs and less reliable in general.

One way to address these limitations is to replace absolute judgments with relative ones, comparing items rather than assessing each in isolation. This idea has roots in the classical law of comparative judgment (Thurstone, 1927). More recently, Shrivastava et al. (2025) apply pairwise comparisons to LLM uncertainty estimation: given two items, the model indicates which one it is more confident about, and the resulting comparisons are aggregated using a Bradley–Terry model. We refer to this formulation as **PairChoice**.

However, PairChoice trades the problems of absolute scoring for a different set of limitations. First, it forces the model to choose one item even when it has similar confidence in both. This can arise, for example, when both items are true or both are false, yet PairChoice still requires a preference. Second, a binary choice loses information about *how much* more confident the model is in one item than the other. For example, confidence scores of (90%, 10%) and (51%, 49%) would induce the same PairChoice outcome despite representing very different confidence gaps.

Motivated by these limitations, we ask a simple

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Method	Context	Output
AbsScore	Single item	A numerical score.
PairChoice	Two items	Binary preference.
PairScore	Two items	Two numerical scores.

Table 1: Comparison of absolute and relative confidence elicitation methods.

question: *rather than choosing between two items, why not score both?* We call this **PairScore**. It retains the pairwise setup of PairChoice, but asks the model to assign a numerical confidence score to each item instead of selecting one over the other (Table 1). PairScore therefore avoids a forced preference while preserving graded confidence information, and its scores can be aggregated directly without fitting a separate ranking model.

To test whether this simple change leads to more useful confidence estimates, we evaluate it across four claim-verification datasets and eight LLMs, as well as best-of- N response selection spanning mathematical reasoning, code generation, knowledge QA, and open-ended instruction following. PairScore generally produces more informative confidence rankings than AbsScore and PairChoice both, although the gains vary across tasks and models. The benefits extend to additional applications: PairScore matches or exceeds AbsScore in 8 of 10 study-selection settings and improves over PairChoice in research-proposal ranking. Finally, we examine when joint scoring helps, studying whether it can recover distinctions when AbsScores collapse and how sensitive its scores are to the choice of paired item.

Our main contributions are:

- We introduce **PairScore**, a simple alternative to PairChoice that retains the pairwise elicitation setup while replacing a forced binary choice with a numerical confidence score for each item. This provides a more flexible form of verbalized LLM confidence elicitation.
- We systematically compare absolute scoring, pairwise choice, and pairwise scoring across claim verification, best-of- N response selection, and additional ranking applications, finding PairScore to be a strong alternative to both AbsScore and PairChoice across diverse models and tasks.
- We study when and how PairScore helps,

showing that joint scoring can recover distinctions obscured by collapsed absolute confidence scores while remaining reasonably robust to the choice of paired item.

2 Related Work

Uncertainty estimation for LLMs. Reliable uncertainty estimation is an active problem, with approaches spanning model log-probability scores, self-consistency across repeated samples (Wang et al., 2023; Kuhn et al., 2023), probing of internal representations (Marks and Tegmark, 2023; Kossen et al., 2024), and direct verbalization (Kadavath et al., 2022). Verbalized confidence is attractive for its simplicity and compatibility with black-box API-only models, but exhibits systematic distortions: scores cluster near scale endpoints (Xiong et al., 2024), shift with prompt phrasing and scale format (Yang et al., 2024), and pursuing better calibration metrics does not eliminate these reliability problems (Xia et al., 2026). Probing work adds a mechanistic angle: uncertainty is linearly encoded in hidden states but does not surface faithfully in numerical outputs (Kirchhof et al., 2025).

Pairwise and relative elicitation. The closest prior work is Shrivastava et al. (2025), who showed that pairwise confidence comparisons aggregated via Bradley–Terry improve selective classification on multiple-choice QA. Their method uses binary preferences over randomly selected pairs. We instead introduce **PairScore**, which retains relative elicitation but asks the model to score both items rather than choose between them. We further evaluate relative elicitation beyond multiple-choice QA, including claim verification and best-of- N selection across several domains. For best-of- N , related approaches use trained pairwise reward models (Liu et al., 2025) or internal log-probabilities (Kang et al., 2025); PairScore is zero-shot and compatible with black-box APIs. Beyond response selection, the AI co-scientist system of Gottweis et al. (2026) uses an Elo-based pairwise tournament to rank research hypotheses, motivating our evaluation of pairwise elicitation for research-proposal ranking. Separately, pairwise LLM judgments have been found to be more stable than pointwise scores for evaluating generated text (Liu et al., 2024b), though this setting concerns output quality rather than model self-confidence.

Comparative judgment and active ranking. The reliability advantage of comparison over absolute rating is classical (Thurstone, 1927) and underlies preference-based LLM training (Christiano et al., 2017; Ouyang et al., 2022). Within active ranking under the Bradley–Terry model, Guo et al. (2018) studied optimal pair selection strategies and Li et al. (2024) applied merge-sort ordering to reduce pairwise evaluation costs; both are relevant to reducing the comparison budget.

3 Methods

We first define the three confidence elicitation methods considered in this work: AbsScore, PairChoice, and PairScore. We then describe how pairwise outputs are aggregated into per-item confidence scores, how pairs are sampled under a fixed query budget, and how the resulting scores are used for ranking and best-of- N selection.

3.1 Score Elicitation

We study three ways of eliciting confidence from an LLM: absolute scoring (**AbsScore**), pairwise choice (**PairChoice**), and our proposed pairwise scoring method (**PairScore**). Full prompts are provided in Appendix B.

AbsScore. The model sees a single item and reports its confidence on a numerical scale from 0 to 100. For example: “Rate your confidence that ‘The Great Wall of China is visible from space’ is factually correct. Output a number from 0–100.” This is a standard approach to verbalized confidence (Kadavath et al., 2022; Xiong et al., 2024).

PairChoice. The model sees two items and chooses the one it is more confident about. For example: “Which statement are you more confident is correct? A: ‘The Great Wall is visible from space.’ B: ‘Water boils at 100°C at sea level.’ Output A or B.” We refer to this approach as PairChoice. Shrivastava et al. (2025) explored this phrasing to find which questions LLM is more confident in answering correctly.

PairScore. PairScore retains the pairwise setup of PairChoice but replaces the forced choice with numerical scoring. Given two items, the model assigns each a confidence score on the same 0–100 scale used by AbsScore. PairScore therefore allows the model to express graded confidence in

both items rather than reducing the comparison to a binary preference.

3.2 Pairwise Aggregation

PairChoice and PairScore both query items in pairs, but require different procedures to obtain a confidence score for each item.

PairChoice. Each comparison produces a binary preference, so we use the Bradley–Terry (BT) model (Bradley and Terry, 1952), following Shrivastava et al. (2025), to aggregate comparisons into per-item ranking scores. Each item i is assigned a latent strength θ_i , with

$$P(i \succ j) = \sigma(\theta_i - \theta_j), \quad (1)$$

where $\sigma(\cdot)$ is the sigmoid function. Given T pairwise comparisons on N items, we estimate the strengths by minimizing the ℓ_2 -regularized negative log-likelihood

$$\min_{\theta \in \mathbb{R}^N} - \sum_{t=1}^T \log \sigma(\theta_{w_t} - \theta_{l_t}) + \frac{\lambda}{2} \|\theta\|_2^2, \quad (2)$$

where w_t and l_t denote the preferred and non-preferred items in comparison t . We optimize this objective using L-BFGS-B and use the fitted strengths θ_i as per-item ranking scores.

PairScore. Each comparison directly returns two numerical confidence scores, $(p_i, p_j) \in [0, 100]^2$. Since an item may appear in multiple comparisons, it can receive multiple scores under different pairings. We use the mean of these scores as its final confidence score. Unlike PairChoice, PairScore therefore does not require fitting an additional ranking model.

Algorithms 1 and 2 summarize the two procedures.

3.3 Pair Sampling and Budget

With N items, exhaustively querying all $\binom{N}{2}$ pairs is generally impractical. We therefore sample pairs uniformly at random under a fixed query budget. For claim verification, all methods receive the same budget of $5N$ LLM calls. AbsScore scores each item five times—once at temperature 0 and four times at temperature 0.7—and averages the resulting scores. PairChoice and PairScore instead use the $5N$ calls for pairwise queries. The comparison is matched by the number of LLM calls rather than by input tokens: a pairwise prompt contains two items and therefore generally requires more input tokens than an AbsScore prompt.

Algorithm 1 PairChoice with Bradley-Terry

Require: Items $x_1, \dots, x_n$; budget B ; regularization λ

Ensure: BT parameters $\theta_1, \dots, \theta_n$

```
1:  $\theta_i \leftarrow 0$  for all  $i$   $\triangleright$  uniform initialization
2:  $\mathcal{O} \leftarrow \emptyset; \mathcal{C} \leftarrow \emptyset$   $\triangleright$  outcomes, compared pairs
3: for  $t = 1$  to  $B$  do
4:    $(i, j) \leftarrow \text{random pair} \notin \mathcal{C}$ 
5:    $\mathcal{C} \leftarrow \mathcal{C} \cup \{(i, j)\}$ 
6:    $w \leftarrow \text{PAIRCHOICELLM}(x_i, x_j)$   $\triangleright$  returns winner, position-debiased
7:    $\mathcal{O} \leftarrow \mathcal{O} \cup \{(w, l)\}$   $\triangleright w$ : winner index,  $l$ : loser index.
8:    $\theta \leftarrow \text{FitBT}(\mathcal{O}, \lambda)$   $\triangleright$  L2-regularized BT
9: end for
10: return  $\sigma(\theta_i)$  as confidence scores  $\triangleright \sigma = \text{sigmoid}$ 
```

Algorithm 2 PairScore

Require: Items $x_1, \dots, x_n$; budget B

Ensure: Confidence scores $s_1, \dots, s_n$

```
1:  $\text{scores}_i \leftarrow []$  for all  $i$   $\triangleright$  collected scores per item
2:  $\mathcal{C} \leftarrow \emptyset$   $\triangleright$  compared pairs
3: for  $t = 1$  to  $B$  do
4:    $(i, j) \leftarrow \text{random pair} \notin \mathcal{C}$ 
5:    $\mathcal{C} \leftarrow \mathcal{C} \cup \{(i, j)\}$ 
6:    $(p_i, p_j) \leftarrow \text{PAIRSCORELLM}(x_i, x_j)$   $\triangleright$  joint scores on 0–100
7:   Append  $p_i$  to  $\text{scores}_i$ ; append  $p_j$  to  $\text{scores}_j$ 
8: end for
9: return  $s_i = \text{mean}(\text{scores}_i)$  as confidence scores
```

3.4 Using the Elicited Scores

Claim verification. Given N items with binary ground-truth, we rank them according to the aggregated confidence scores. We evaluate this ranking using AUC-ROC; higher AUC-ROC implies better separation of positive and negative items.

Best-of- N selection. Given K candidate responses to a problem, the goal is to select the best response. AbsScore scores each candidate independently and selects the response with the highest confidence. For PairChoice and PairScore, we use sequential elimination procedures (**PC-Elim** and **PS-Elim**) requiring $K - 1$ pairwise comparisons. At each comparison, PC-Elim advances the response selected by PairChoice, while PS-Elim ad-

vances the response receiving the higher PairScore. For tasks with directly matchable answers, we additionally compare against **Majority Vote**, which selects the most frequent answer.

Additional ranking tasks. We also evaluate the elicited scores on two additional ranking problems. For systematic-review study selection, abstracts are ranked by the model’s confidence that they satisfy the review’s inclusion criteria, and we evaluate AUC-ROC against the final inclusion decisions. For research-proposal ranking, pairwise model judgments are compared against expert quality scores. Full setup and results are provided in Appendix D.

For claim verification and best-of- N selection, confidence is elicited from the same LLM being evaluated; no separate judge model is used for confidence-based selection.

4 Experiments

We evaluate whether pairwise scoring provides more useful confidence estimates than absolute scoring and pairwise choice across two main applications: claim verification (Section 4.1) and best-of- N response selection (Section 4.2). We then test whether these gains extend to additional ranking tasks, and analyze two aspects of PairScore’s behavior: whether joint scoring can recover distinctions when absolute scores collapse and how sensitive the resulting scores are to the paired item (Section 4.4).

4.1 Claim Verification

Setup. We evaluate on four binary claim-verification datasets: CREAK (Onoe et al., 2021) (entity facts), FEVER (Thorne et al., 2018) (Wikipedia), TruthfulQA (Lin et al., 2022) (adversarial QA), and CliniFact (Zhang et al., 2025) (clinical claims). We use $N=500$ items from each dataset with a fixed seed and report AUC-ROC against the ground-truth binary labels. We evaluate eight models spanning different families and sizes, giving 32 model–dataset settings. Dataset and subsampling details are provided in Appendix C.1, and exact model and inference details in Appendix C.2.

Does relative elicitation improve over absolute scoring? Relative elicitation improves confidence ranking in a substantial fraction of the model–dataset settings, although the magnitude of the gain

varies (Table 2). Relative methods are particularly helpful for some of the smaller models: for Qwen3-1.7B, the best pairwise result improves AUC-ROC from 78.4 to 83.4 on CREAK and from 63.1 to 71.7 on FEVER. A similar pattern appears for Qwen3-4B, with improvements from 81.7 to 86.1 on FEVER and from 73.1 to 79.1 on TruthfulQA. The gains are smaller when AbsScore already provides a strong ranking; for example, on FEVER, GPT-5-mini and Gemini-2.5-FL achieve 95.6 and 91.4 AUC-ROC with AbsScore(5 $\times$), respectively, with no improvement from relative elicitation. Full results across all eight models and four datasets are reported in Table 2.

Does PairScore improve over PairChoice?

PairScore outperforms PairChoice in 24 of the 32 model–dataset combinations (Table 2), with the clearest pattern on TruthfulQA, where it performs better for 7 of the 8 models. Since the two methods use the same paired inputs and query budget, this comparison isolates the form of the elicited judgment: PairChoice reduces each comparison to a binary preference, whereas PairScore retains a graded confidence estimate for both items. The results indicate that this additional information is often useful for confidence ranking, although PairChoice remains competitive in some settings.

How does performance vary with the query budget? The preceding comparisons use the full $5N$ -call budget. We next examine how performance changes as pairwise comparisons are added. Figure 1 shows the resulting AUC-ROC curves for GPT-5-mini and Qwen3-4B, with AbsScore(1 $\times$) and AbsScore(5 $\times$) as reference points. PairScore generally improves steadily with additional comparisons and often reaches or exceeds the AbsScore(5 $\times$) baseline before using the full budget. PairChoice converges less consistently and frequently remains below the same baseline. The corresponding curves for all eight models are reported in Appendix C.3; a complementary comparison based on token usage is given in Appendix C.5.

4.2 Best-of- N Selection

We next ask whether the elicited confidence scores are useful for selecting among multiple candidate responses. For each problem, we generate $N=8$ responses at temperature 0.7 and use each method to select one response.

Setup. We compare Random selection, AbsScore, PairChoice-Elim (PC-Elim), PairScore-Elim (PS-Elim), and an Oracle upper bound. AbsScore scores each candidate independently and selects the highest-scoring response, while PC-Elim and PS-Elim use $N-1$ pairwise comparisons to select among the candidates. All confidence-based selection is performed by the same model that generated the responses; no external model is used for selection.

Our main evaluation spans seven benchmarks covering mathematical reasoning, code generation, knowledge QA, and open-ended instruction following. We evaluate GPT-4o-mini, GPT-5-mini, and Gemini-2.5-Flash-Lite; results for GPT-4.1-nano are reported in Appendix C.4. Full benchmark details are given in Table 7. For mathematical reasoning, where final answers can also be matched directly, we separately compare against majority-vote baselines below.

When does pairwise selection help? The benefit of pairwise selection is strongly task-dependent (Table 4). The clearest gains appear on HumanEval+ and AlpacaEval. On HumanEval+, a pairwise method outperforms AbsScore for each of the three generator models, with particularly large gains for Gemini-2.5-Flash-Lite. On AlpacaEval, pairwise selection also improves the average judge score for all three models; moreover, when AbsScore and a pairwise method select different responses, the pairwise-selected response is preferred in 70–88% of the non-tied comparisons (Table 3). In contrast, the methods are nearly indistinguishable on MBPP+, while results on mathematical reasoning and knowledge QA are more mixed.

What happens when answers can be directly matched?

For mathematical reasoning, the candidate responses provide an additional signal: their final answers can be matched directly. We therefore compare confidence-based selection against Majority Vote (MV) and confidence-weighted variants (Table 10). In MV, responses vote for their final answer and the most frequent answer is selected; in weighted MV, these votes are weighted using either AbsScore (wMV-AbsS) or PairChoice-Elim confidence (wMV-PCe). Across the math benchmarks, majority voting is generally the strongest non-oracle baseline, while confidence weighting provides little additional benefit. Pairwise selection can still improve over AbsScore on the harder prob-

Table 2: AUC-ROC (%) for all eight models across four claim-verification datasets. **Bold** = best method per model–dataset setting; * = significantly better than AbsScore(5x) at 95% (bootstrap).

Dataset	Method	GPT-5 mini	GPT-4o mini	GPT-4.1 nano	Gemini 2.5-FL	Qwen3 4B	Qwen2.5 3B	Phi-4 mini	Qwen3 1.7B
CREAK	AbsScore(1x)	94.7	95.7	94.6	95.0	88.9	87.7	83.9	71.8
	AbsScore(5x)	97.9	96.5	95.1	96.7	89.7	88.5	89.2	78.4
	PairChoice	97.6	96.1	94.9	93.8	83.2	76.3	89.9	59.0
	PairScore	96.4	97.5*	95.8	97.1	91.9*	90.1	88.1	83.4*
FEVER	AbsScore(1x)	94.9	89.8	85.6	90.1	81.3	79.0	79.9	62.3
	AbsScore(5x)	95.6	91.0	87.9	91.4	81.7	80.4	82.1	63.1
	PairChoice	94.3	90.1	89.8*	85.9	77.7	66.6	85.9*	49.4
	PairScore	93.7	92.3	90.2*	91.1	86.1*	73.7	82.3	71.7*
TruthfulQA	AbsScore(1x)	86.1	87.0	76.3	86.5	73.4	65.3	63.8	56.2
	AbsScore(5x)	91.3	87.9	76.1	87.1	73.1	65.2	65.1	58.6
	PairChoice	94.5*	85.7	74.3	88.5	71.7	72.4*	67.1	57.2
	PairScore	92.0	87.1	75.6	91.7*	79.1*	82.4*	74.7*	69.1*
CliniFact	AbsScore(1x)	64.0	60.1	52.7	60.7	56.4	55.1	52.3	48.8
	AbsScore(5x)	65.0	62.1	53.8	60.4	56.4	57.8	54.6	47.0
	PairChoice	69.6*	58.2	59.5*	59.4	50.4	61.1	52.0	48.4
	PairScore	71.1*	63.4	56.5	54.4	56.5	50.5	55.7	55.8*

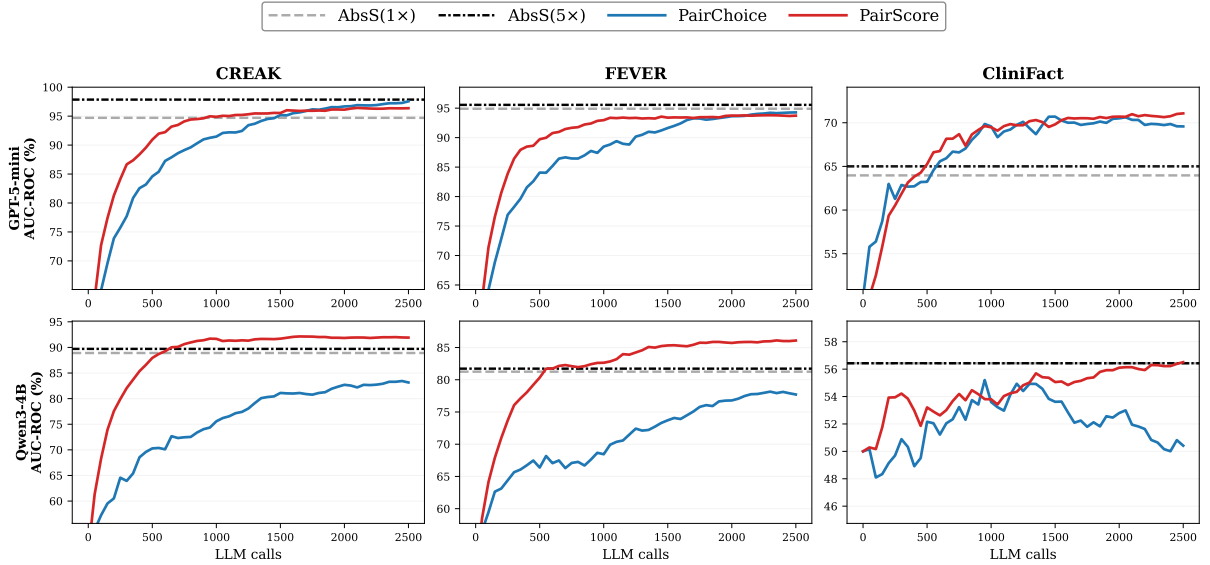


Figure 1: Convergence of AUC-ROC (%) vs. LLM calls for GPT-5-mini (top) and Qwen3-4B (bottom) across CREAK, FEVER, and CliniFact. Gray dashed = AbsScore(1x); black dash-dot = AbsScore(5x). PairScore reaches or exceeds the 5x AbsScore baseline with far fewer calls; PairChoice often does not reach it. Full results for all 8 models in Appendix C.3.

lems, but when answer agreement is available, it provides a weaker signal than directly aggregating the candidate answers.

4.3 Additional Applications

Does PairScore generalize to other ranking problems? We further evaluate PairScore on two additional applications. The first is study selection for five systematic reviews spanning computer science, conservation, medicine, biology, and ecology

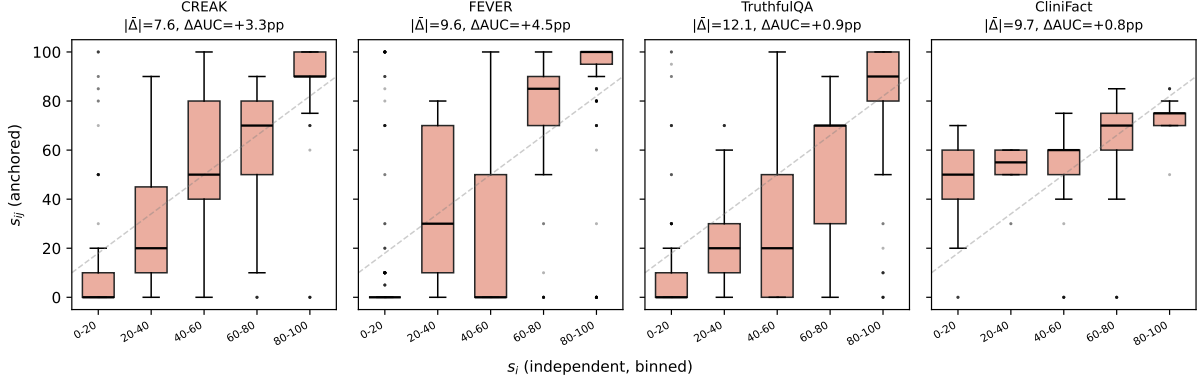


Figure 2: Joint scores vs. independent AbsScores, by dataset (GPT-4o-mini, 500 pairs per dataset). Items are grouped by their independent AbsScore (x-axis); boxes show the distribution of scores the same items receive when scored jointly with a partner. Wide boxes indicate that joint scoring spreads out items that AbsScore assigns nearly identical scores; the annotated mean absolute shift and AUC change are given per dataset (AUC improves on all four).

Table 3: Head-to-head: AbsScore vs. pairwise (AlpacaEval, $n=200$, GPT-4o-mini judge).

Generator	Method	AbsS wins	Method wins	Tie
GPT-4o-mini	PC-Elim	6.5%	15.5%	78.0%
	PS-Elim	5.0%	22.5%	72.5%
GPT-5-mini	PC-Elim	4.0%	23.0%	73.0%
	PS-Elim	6.0%	23.0%	71.0%
Gemini-2.5-FL	PC-Elim	5.0%	18.5%	76.5%
	PS-Elim	2.5%	19.0%	78.5%

(De Bruin et al., 2023; Hall et al., 2012; Walker et al., 2018; Leenaars et al., 2020; Bladon et al., 2022; Chausson et al., 2020). PairScore matches or exceeds AbsScore in 8/10 settings, compared with 3/10 for PairChoice (Table 5). Second, on research-proposal ranking, we evaluate 656 within-topic comparisons among 93 proposals with expert quality scores (Si et al., 2024). PairScore improves pairwise ranking accuracy from 50.6% with PairChoice to 60.7%. These results show that the benefits of joint numerical scoring also extend to ranking problems beyond claim verification and best-of- N selection. Full experimental details are provided in Appendix D.

4.4 Understanding PairScore

A recurring limitation of AbsScore is *score collapse*: the model assigns identical or near-identical confidence scores to items that differ in correctness or quality. In claim verification, true and false claims can therefore receive the same score; in best-of- N selection, several candidate responses may

all receive near-ceiling confidence. We examine whether joint scoring can distinguish items that AbsScore fails to separate, and whether the resulting scores are sensitive to the confidence level of the paired item. Concrete examples of score collapse are provided in Appendix C.6.

Does joint scoring alleviate score collapse? We compare the score assigned to an item in isolation with the score assigned to the same item when evaluated jointly with another item. Using GPT-4o-mini, we score 500 pairs per dataset both independently and jointly. Figure 2 shows that joint scoring can substantially spread out scores that are concentrated near the extremes or midpoint under AbsScore. These changes are also useful for ranking: the jointly elicited scores achieve higher AUC-ROC than the corresponding independent scores on all four datasets. Thus, joint scoring can distinguish between items that receive the same or very similar absolute confidence scores.

How sensitive is PairScore to the paired item?

The previous experiment also raises a potential concern: the score assigned to an item may change depending on its comparison partner. We test this by pairing each item with anchors drawn from low-, medium-, and high-confidence groups (Table 13; GPT-4o-mini, 100 items per dataset, 10 anchors per confidence level). PairScore changes somewhat across anchor groups, but the differences between low-, medium-, and high-confidence anchors are generally modest relative to the variation across items. Thus, while PairScore is not invariant to the paired item, we find limited sensitivity to the

Table 4: Best-of-8 selection. Accuracy (%) for math, code, and knowledge QA; average GPT-4o-mini judge score (0–7 scale) for AlpacaEval (open-ended instruction following). AbsScore rates each response once on 0–100 and picks the highest score (single call per response; no averaging over repetitions, unlike claim verification). **Bold** = best non-oracle per (model, dataset). HuEval+ = HumanEval+. GPT-4.1-nano results in Appendix C.4.

Model	Method	Math		Code		Knowledge QA		Open
		Math 500	AIME	HuEval+	MBPP+	GPQA	Trivia QA	Alpaca
GPT-4o-mini	Random	66.8	8.9	64.6	45.4	29.3	64.0	3.52
	AbsScore	71.2	11.1	68.3	46.6	33.3	66.0	3.88
	PC-Elim	72.0	8.9	67.7	45.4	34.3	71.0	4.19
	PS-Elim	71.8	12.2	69.5	46.2	33.3	67.0	4.43
	Oracle	82.2	14.4	76.2	51.4	71.2	89.0	6.18
GPT-5-mini	Random	74.0	40.0	75.0	51.5	64.1	85.5	3.68
	AbsScore	75.5	46.7	73.8	51.0	63.1	85.5	3.96
	PC-Elim	76.5	50.0	81.1	51.0	64.1	81.0	4.96
	PS-Elim	75.5	46.7	79.9	51.5	58.1	86.0	4.52
	Oracle	82.5	56.7	84.8	54.5	86.4	92.5	6.36
Gemini-2.5-FL	Random	54.0	7.8	61.6	52.0	28.8	63.5	3.81
	AbsScore	55.0	11.1	57.3	50.0	32.8	64.5	3.50
	PC-Elim	58.0	15.6	72.0	51.5	33.8	65.5	3.98
	PS-Elim	55.5	13.3	69.5	50.0	30.8	62.0	4.56
	Oracle	65.5	18.9	80.5	57.5	66.7	87.0	6.01

Table 5: External study-selection results, AUC-ROC (%). Bold marks the best elicitation method in each row. PairScore matches or exceeds AbsScore in 8/10 settings, compared with 3/10 for PairChoice.

Review — domain	Model	AbsScore	PairChoice	PairScore
Hall et al. (2012) — computer science	Gemini-2.5-FL	98.2	96.9	98.4
	GPT-4o-mini	97.8	98.0	99.1
Bladon et al. (2022) — conservation	Gemini-2.5-FL	93.8	91.8	92.9
	GPT-4o-mini	96.7	95.6	96.0
Walker et al. (2018) — biology & medicine	Gemini-2.5-FL	83.5	79.8	85.1
	GPT-4o-mini	79.1	83.1	83.1
Leenaars et al. (2020) — medicine	Gemini-2.5-FL	76.1	81.9	78.8
	GPT-4o-mini	78.6	76.0	78.6
Chausson et al. (2020) — ecology	Gemini-2.5-FL	85.1	79.2	85.7
	GPT-4o-mini	82.7	78.8	83.3

partner’s confidence level.

5 Conclusion

We introduced **PairScore**, which elicits confidence by scoring two items jointly rather than independently (AbsScore) or through a forced choice (PairChoice). Across our experiments, PairScore often improves confidence ranking and selection, with particularly clear benefits in settings where absolute scores provide limited discrimination. At the same time, the gains are task-dependent, and sim-

pler alternatives such as PairChoice or majority voting remain competitive in some settings. Overall, our results suggest that joint numerical scoring offers a simple and useful middle ground between absolute scores and binary comparison.

Limitations and Future Work. We do not yet understand why joint scoring changes confidence estimates. Future work could focus on mechanistic understanding of PairScore and relative confidence estimation in general. Further, the token usage could be optimized with better pair selection.

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A Algorithm Pseudocode

Algorithm 3 AbsScore (Repeated Numerical Scoring)

Require: Items $x_1, \dots, x_n$; repetitions R ; temperature τ

Ensure: Confidence scores $s_1, \dots, s_n$

```

1: for  $i = 1$  to  $n$  do
2:    $s_i^{(1)} \leftarrow \text{ABSSCORELLM}(x_i, \text{temp}=0)$   $\triangleright$ 
     deterministic
3: end for
4: for  $r = 2$  to  $R$  do
5:   for  $i = 1$  to  $n$  do
6:      $s_i^{(r)} \leftarrow \text{ABSSCORELLM}(x_i, \text{temp}=\tau)$ 
7:   end for
8: end for
9: return  $s_i = \frac{1}{R} \sum_{r=1}^R s_i^{(r)}$  [ $nR$  calls]

```

B Prompt Templates

AbsScore.

How confident are you that the following statement is correct.

Statement: “{text}”

Provide a single number between 0 (certainly wrong) and 100 (certainly correct). Output ONLY the number.

PairChoice.

Which of the following statements are you more confident of being correct?

A: “{text_a}”

B: “{text_b}”

Output ONLY ‘A’ or ‘B’.

PairScore.

How confident are you that each of the following statements is factually correct? Provide a score on a scale of 0 to 100.

A: “{text_a}”

B: “{text_b}”

Output format: “A: [score], B: [score]”

Best-of- N : AbsScore. For verifiable tasks (math, code, knowledge QA):

Here is a {task label} and a proposed answer.

Question: {question}

{Answer/Solution}: {response}

How confident are you that {quality condition}? Output a single integer between 0 (certainly wrong) and 100 (certainly correct). Output ONLY the number.

For open-ended instruction following:

Here is an instruction and a response. Rate the response quality as a percentage from 0 to 100.

Instruction: {question}

Response: {response}

Consider: accuracy, completeness, clarity, helpfulness, and whether it fully addresses the instruction. Be critical – most responses should score between 30 and 80. A score above 90 means near-perfect.

Output ONLY a single integer (0-100).

Best-of- N : PairChoice.

Here is a {task label} and two proposed answers.

Question: {question}

{Answer/Solution} A: {response_a}

{Answer/Solution} B: {response_b}

Which answer is {quality condition}?

Output ONLY ‘A’ or ‘B’.

For open-ended tasks the final question is “Which response better answers the instruction?”.

Best-of- N : PairScore.

Here is a {task label} and two proposed answers.
 Question: {question}
 {Answer/Solution} A: {response_a}
 {Answer/Solution} B: {response_b}
 How confident are you that each answer is correct on a scale of 0 to 100.
 Output format: “A: [score], B: [score]”

For open-ended tasks the scoring instruction is “Rate the quality of each response on a scale of 0 to 100. Consider accuracy, completeness, clarity, and helpfulness.”

The {task label}, {Answer/Solution}, and {quality condition} slots are task-dependent, e.g., “math problem” / “Solution” / “more likely to have the correct final answer” for math, and “programming problem” / “Solution” / “this code is correct and handles all edge cases” for code.

AlpacaEval judge (response scoring). The judge score of a response is its number of wins in pairwise comparisons against the other $N-1$ candidates (0–7 for $N=8$), with presentation order randomized per comparison:

Compare two responses to the same instruction. Consider accuracy, helpfulness, detail, and clarity.
 Instruction: {instruction}
 Response A: {response_a}
 Response B: {response_b}
 Which response is better? Output ONLY ‘A’ or ‘B’.

AlpacaEval judge (head-to-head). For the head-to-head evaluation (Table 3), the judge compares the two selected responses in both orderings; agreement gives a win, disagreement counts as a tie:

Here is an instruction and two responses. Which response better answers the instruction? Consider accuracy, helpfulness, detail, and clarity.
 Instruction: {instruction}
 Response A: {response_a}
 Response B: {response_b}
 Output ONLY ‘A’, ‘B’, or ‘Tie’.

C Additional Experimental Results

C.1 Datasets

Claim verification. Table 6 lists the four claim-verification datasets. Each is subsampled to $n=500$ items using a fixed random seed (seed 42), stratified to preserve the original class ratio where possible.

Table 6: Claim-verification datasets. T/F = True/False label counts in our 500-item subsample.

Dataset	Domain	T	F
CREAK (Onoe et al., 2021)	Entity facts	242	258
FEVER (Thorne et al., 2018)	Wikipedia	252	248
TruthfulQA (Lin et al., 2022)	Adversarial QA	96	404
CliniFact (Zhang et al., 2025)	Clinical claims	140	360

Best-of- N . Table 7 lists the eight best-of- N benchmarks. For benchmarks larger than our target n , we take the first n problems in the canonical ordering (no random subsampling, to allow reproducibility).

Table 7: Best-of- N datasets. All use $N=8$ responses per problem.

Dataset	Category	n
GSM8K (Cobbe et al., 2021)	Math reasoning	200
MATH-500 (Hendrycks et al., 2021)	Math competition	200
AIME 2024	Math olympiad	90
HumanEval+ (Liu et al., 2024a)	Code generation	164
MBPP+ (Austin et al., 2021)	Code generation	200
GPQA Diamond (Rein et al., 2023)	Science QA	198
TriviaQA (Joshi et al., 2017)	Factual QA	200
AlpacaEval (Dubois et al., 2024)	Open-ended	200

C.2 Model Details

API models. Table 8 lists the exact API identifiers used via OpenRouter.

Table 8: API models and identifiers. FL = Flash-Lite.

Label	API identifier	Used for
GPT-5-mini	openai/gpt-5-mini	Claim verif + BoN
GPT-4o-mini	openai/gpt-4o-mini	Claim verif + BoN
GPT-4.1-nano	openai/gpt-4.1-nano	Claim verif + BoN
Gemini-2.5-FL	google/gemini-2.5-flash-lite	Claim verif + BoN

All API calls were made through OpenRouter (<https://openrouter.ai>). For AbsScore, we use temperature 0.0 for the deterministic call and temperature 0.7 for the four stochastic repetitions. For pairwise methods, all comparisons use temperature 0.0 and each comparison is a single call. The order of the two items in the prompt is randomized independently for each comparison to debias position effects. For PairChoice, outputs that name neither item are resolved by a random choice; for PairScore, scores are mapped back to the original

item order.

Local models. Table 9 lists the local models. All local models were run in bfloat16 precision on a single NVIDIA RTX 2080 Ti (11 GB), using the HuggingFace transformers library without quantization. Inference used greedy decoding (temperature 0.0) for all local model experiments. No VLLM was used; all inference is single-GPU autoregressive with batch size 1.

Table 9: Local models. All run in bfloat16 on one RTX 2080 Ti.

Label	HuggingFace ID	Params
Qwen3-4B	Qwen/Qwen3-4B	4B
Qwen2.5-3B	Qwen/Qwen2.5-3B-Instruct	3B
Phi-4-mini	microsoft/Phi-4-mini-instruct	3.8B
Qwen3-1.7B	Qwen/Qwen3-1.7B	1.7B

C.3 Convergence: All Models

Figures 3–6 show AUC-ROC (%) vs. LLM calls for all 8 models on each dataset (4 models per row, 2 rows per figure). The pattern is consistent across model families: PairScore typically reaches or exceeds the AbsScore($5\times$) baseline within 500–1000 calls, with the largest gains for the weakest models (Qwen3-1.7B); PairChoice does so less often.

Note: The main experiments in Section 4 use $N=500$ items per dataset, 8 models, and the $[0, 100]$ scale with AbsScore, PairChoice, and PairScore. Full per-dataset AUC-ROC tables are in Appendix C.7. The consistency, calibration, and distribution analyses below were conducted on earlier $N=250$ runs with the $[0, 1]$ scale and partially different method sets; the qualitative conclusions remain valid.

C.4 Best-of-8: GPT-4.1-nano Results

Table 11 shows best-of-8 results for GPT-4.1-nano. The pattern mirrors the stronger models: pairwise methods gain most on code and open-ended tasks. AIME was not evaluated for nano.

C.5 Token Efficiency

Table 12 reports, for each dataset and pairwise method, the fraction of (dataset, model) settings where the method eventually exceeds AbsScore($5\times$) AUC, and the mean token count at first crossover as a fraction of the AbsScore($5\times$)

token budget. Token counts for numerical calls are exact (from prompt_tokens + completion_tokens in the call log); pairwise token counts are estimated via tiktoken (cl100k_base) from the item texts and prompt templates, with fixed completion token budgets (2 for PairChoice, 12 for PairScore).

The key takeaway: despite consuming $\sim 1.0\times$ (PairChoice) or $\sim 1.4\times$ (PairScore) the tokens of AbsScore($5\times$) in total, pairwise methods reach the AbsScore($5\times$) threshold at a fraction of those tokens when they do exceed it. PairScore is most consistent, exceeding the AbsScore($5\times$) AUC in 74% of settings at $0.65\times$ of its token budget.

C.6 Anecdotal Examples

Tables 14 and 15–17 show concrete cases illustrating when pairwise methods succeed where AbsScore fails.

Claim verification. Table 14 shows pairs where AbsScore assigns identical scores to a true and false claim, but PairScore correctly separates them. A recurring pattern is *floor collapse*: AbsScore outputs 0 for both items (the model declines to commit to either), but when forced to compare, it correctly identifies the true claim. A second pattern is *midpoint collapse*: AbsScore outputs 50 for both items (maximum uncertainty), but the comparison resolves the ambiguity.

Best-of- N . Tables 15–17 show three best-of-8 examples on AlpacaEval (GPT-5-mini generator), each illustrating a different failure mode of AbsScore.

Example 1: Explanation quality. The task is to classify a LinkedIn invitation email as Promotions or Social. All 8 responses give the correct label, but differ in whether they justify it. AbsScore assigns 100 to two responses and picks [2], which gives the bare label with no explanation (judge = 0). Both pairwise methods select responses with clear justifications (judge = 6).

Example 2: Attribution. The task is to complete the aphorism “two things are infinite...” All 8 responses produce the correct quote, but only [5] includes the attribution to Albert Einstein. AbsScore assigns 100 to all responses and picks [0] (no attribution; judge = 3). Both pairwise methods select [5] (judge = 7).

Example 3: Completeness. The task is to state

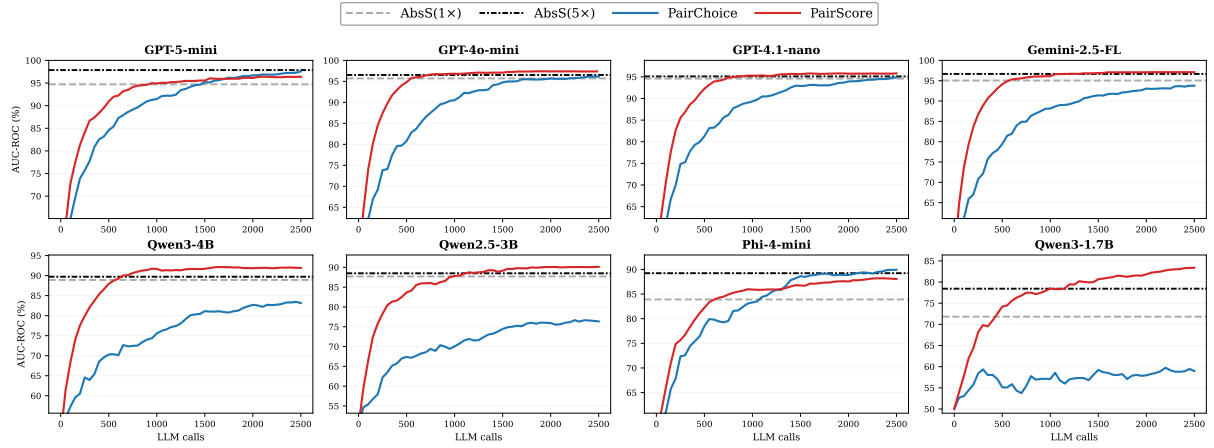


Figure 3: Convergence on CREAK. All 8 models; gray dashed = AbsScore(1 $\times$); black dash-dot = AbsScore(5 $\times$).

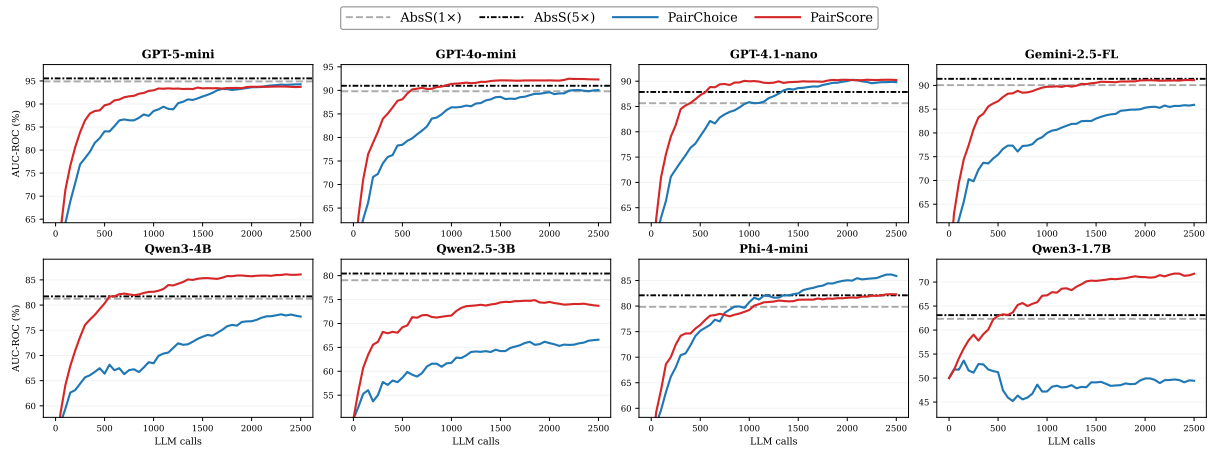


Figure 4: Convergence on FEVER.

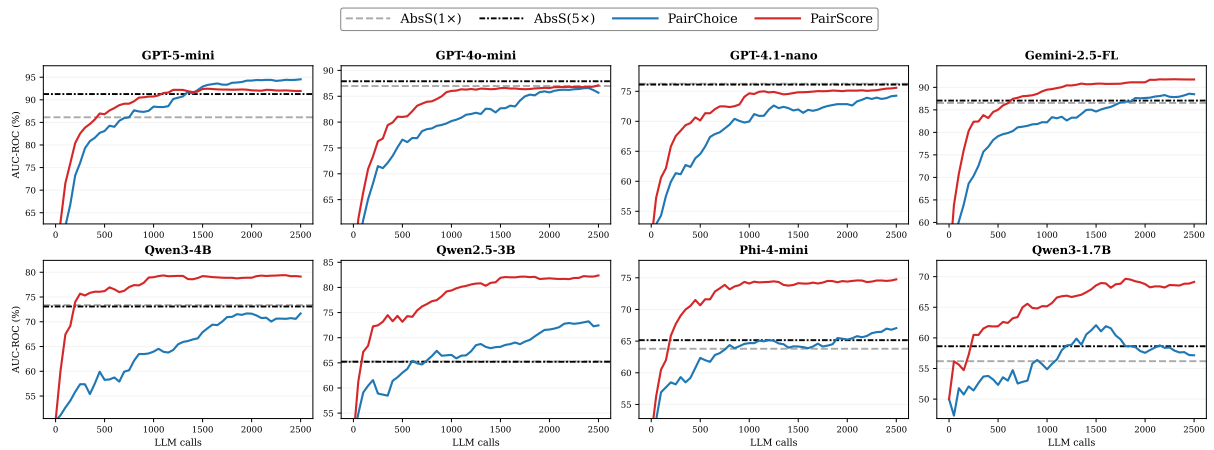


Figure 5: Convergence on TruthfulQA.

when the Yamato battleship was built. AbsScore assigns 100 to two responses and picks [0]: “The Yamato Battleship was built in 1941,” which gives only the commissioning year (judge = 1). The more complete answer distinguishes the laying-down date (1937) from the commissioning date

(1941); PairScore assigns it the highest mean score and PC-Elim selects it (judge = 5 or 7).

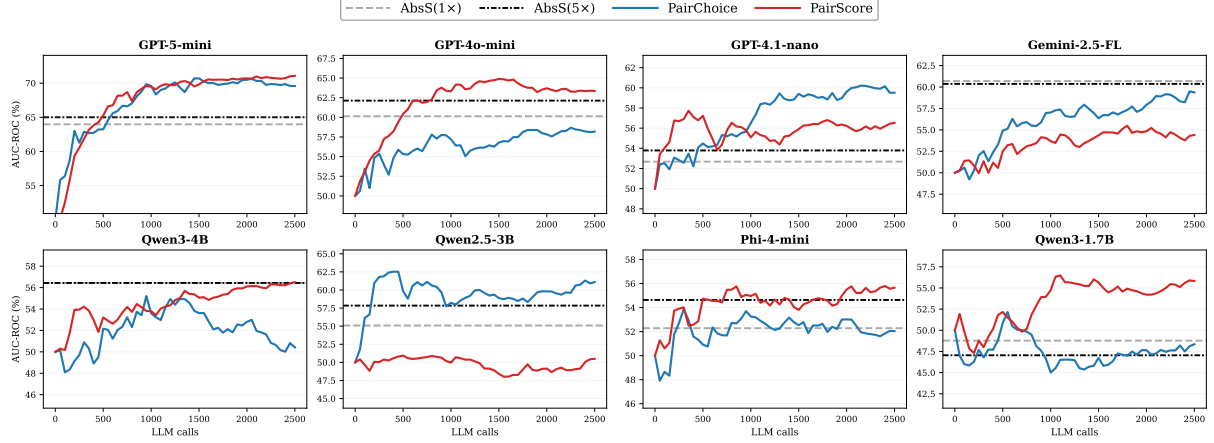


Figure 6: Convergence on CliniFact.

Table 10: Best-of-8 accuracy (%) on math with Majority Vote. AbsS = AbsScore; PCe = PC-Elim; PSe = PS-Elim; Orac = Oracle (best of 8, upper bound); wMV-AbsS/PCe = weighted MV using AbsScore/PC-Elim confidence weights. **Bold** = best non-oracle.

Dataset	Model	Rand	MV	AbsS	PCe	PSe	wMV AbsS	wMV PCe	Orac
GSM8K	GPT-4.1-nano	92.0	95.5	92.0	92.0	91.5	95.5	94.0	97.5
	GPT-4o-mini	91.5	90.0	89.0	88.5	89.0	89.5	89.5	96.0
	Gemini-2.5-FL	87.5	91.5	89.0	87.5	87.0	91.0	91.0	93.0
MATH-500	GPT-4.1-nano	59.0	71.5	64.0	64.0	66.0	72.0	70.0	75.0
	GPT-4o-mini	67.5	71.5	69.5	72.0	68.5	71.5	70.0	80.5
	GPT-5-mini	74.0	76.0	75.5	76.5	75.5	76.0	76.5	82.5
AIME	GPT-5-mini	40.0	56.7	46.7	50.0	46.7	55.6	56.7	56.7

Table 11: Best-of-8 selection for GPT-4.1-nano. Accuracy (%) for math/code/QA; avg. judge score (0–7) for AlpacaEval. AIME not evaluated for this model. **Bold** = best non-oracle.

Method	Math	Code		Knowledge QA		Open
	Math 500	HuEval+	MBPP+	GPQA	Trivia QA	Alpaca
Random	57.8	73.8	48.0	43.4	80.0	3.38
AbsScore	59.8	72.0	46.4	52.0	78.0	3.77
PC-Elim	61.2	75.0	47.2	47.5	81.0	4.38
PS-Elim	62.6	75.0	46.2	45.5	81.0	4.70
Oracle	73.4	81.1	51.8	85.4	92.0	6.38

C.7 Full Per-Model AUC-ROC Results

Table 18 reports AUC-ROC (%) for all eight models across all four datasets, including the three shown in the main paper (Table 2) for completeness. **Bold** = best per row; * = significantly better than AbsScore(5x) at 95% (bootstrap over items). AbsScore(1x)/AbsScore(5x) = single/5-averaged AbsScore.

D Study Selection and Research-Proposal Ranking

We evaluate the same elicitation methods on two external applications:

1. **Study selection for systematic reviews.** We evaluate title-and-abstract screening, the high-volume first stage of study selection, in which reviewers exclude irrelevant records before full-text assessment. We use Hall, Walker, and Leenaars from the SYNERGY study-selection benchmark (De Bruin et al., 2023; Hall et al., 2012; Walker et al., 2018; Leenaars et al., 2020), the Bladon butterfly-and-moth conservation synopsis (Bladon et al., 2022), and the Chausson Nature-based Solutions review in ecology (Chausson et al., 2020). The labels indicate final inclusion in the corresponding evidence synthesis. Each method receives a budget of $5N$ calls: BinaryScore and AbsScore query each abstract five times, while PairChoice and

Table 12: Token efficiency of pairwise methods relative to AbsScore($5\times$). “Exceeds” = fraction of (dataset, model) settings where the method eventually surpasses AbsScore($5\times$) AUC. “Token ratio” = mean tokens at first crossover $\div$ AbsScore($5\times$) token budget (lower is more efficient; entries where the method never exceeds are counted at total token usage). Settings without complete token logs are excluded, leaving 27 of 32 settings.

Dataset	Method	Exceeds AbsS($5\times$)		Token ratio at crossover	
		Count	%	Mean	(vs. AbsS)
CREAK	PairChoice	1/6	17%	134K	0.82 $\times$
	PairScore	5/6	83%	101K	0.64 $\times$
FEVER	PairChoice	2/7	29%	116K	0.72 $\times$
	PairScore	5/7	71%	116K	0.73 $\times$
TruthfulQA	PairChoice	3/7	43%	151K	0.81 $\times$
	PairScore	5/7	71%	94K	0.53 $\times$
CliniFact	PairChoice	3/7	43%	185K	0.79 $\times$
	PairScore	5/7	71%	178K	0.71 $\times$
Overall	PairChoice	9/27	33%	147K	0.78 $\times$
	PairScore	20/27	74%	123K	0.65 $\times$

Table 13: Mean PairScore ($\pm$ std. dev.) when items are paired with low-, medium-, and high-confidence anchors (GPT-4o-mini). Shifts across anchor levels are small relative to the differences between items.

Dataset	AbsScore bin	n	AbsS	PS(low)	PS(med)	PS(high)
CREAK	0–20	49	2.7	9.8 $\pm$ 15.1	12.1 $\pm$ 13.0	11.2 $\pm$ 13.9
	60–80	4	71.2	74.2 $\pm$ 4.0	73.4 $\pm$ 5.4	71.0 $\pm$ 6.5
	80–100	40	93.1	91.3 $\pm$ 5.9	91.1 $\pm$ 5.3	92.2 $\pm$ 6.1
TruthfulQA	0–20	48	3.5	17.7 $\pm$ 15.7	17.2 $\pm$ 13.2	19.6 $\pm$ 15.5
	20–40	20	26.0	43.5 $\pm$ 17.6	42.4 $\pm$ 16.6	43.5 $\pm$ 14.8
	80–100	20	89.5	85.3 $\pm$ 7.0	84.0 $\pm$ 13.6	85.7 $\pm$ 11.2
CliniFact	40–60	60	50.0	56.1 $\pm$ 8.4	62.5 $\pm$ 6.8	64.5 $\pm$ 6.1
	60–80	38	70.4	68.2 $\pm$ 6.2	72.2 $\pm$ 4.5	72.9 $\pm$ 3.6

Table 14: Claim verification examples where AbsScore ties but PairScore correctly separates (GPT-4o-mini). AbsS = AbsScore($5\times$) score; PS = PairScore.

True claim	False claim	AbsS (T/F)	PS (T/F)
Nancy Pelosi followed Paul Ryan as Speaker of The House.	Richard Linklater directed <i>Gone with the Wind</i> .	0 / 0	84 / 0
Annie is a movie.	Benjamin Franklin was born in 1790.	0 / 0	96 / 0
ZX008 0.5 mg/kg/day is superior to placebo in reducing convulsive seizures [correct clinical claim].	Umooze is superior to placebo on QoL index [incorrect clinical claim].	50 / 50	74 / 8

PairScore sample $5N$ unique pairs uniformly without replacement. PairChoice outcomes are aggregated with a regularized Bradley–Terry model and PairScore with the per-item mean. The abstract sample is fixed and A/B order is randomized independently. Table 5 reports the saved $N = 500$ run for each API model (Bladon

uses its full reconstructed corpus, $N = 457$).

Binary-to-numerical expression is the most consistent change: AbsScore exceeds BinaryScore in all ten settings. PairScore has smaller losses than PairChoice when pairing does not help, while retaining gains where AbsScore leaves headroom. For example, PairScore im-

Table 15: AlpacaEval example 1 (GPT-5-mini). **Prompt:** “Decide if the given email belongs to the Promotions or Social category. Subject: You have an invitation. Email: Hi, I’d like to join your LinkedIn network.” AbsS scores are 95 except where shown. Judge scores on 0–7.

#	Response	AbsS	PS	Judge
0	Social. It’s a LinkedIn connection request, not a promotional offer.	95	94.3	5
1	Social	95	84.3	2
2	Social	100	89.3	0 ← AbsS
3	Social	95	87.9	1
4	Social	95	87.1	3
5	Social. It’s a LinkedIn connection invitation, not a promotional message.	100	95.0	6
6	Social. It’s a LinkedIn connection request: a personal networking invitation, not a promotion.	95	95.0	6 ← PCe
7	Social. It’s a LinkedIn network invitation (personal/social networking), not a promotional message.	95	95.3	5 ← PSe

Table 16: AlpacaEval example 2 (Gemini-2.5-FL). **Prompt:** “Complete the aphorism and write the original quote as the answer. two things are infinite” All AbsS scores are 100. Judge scores on 0–7.

#	Response	PS	Judge
0	“two things are infinite: the universe and human stupidity; and I’m not sure about the universe.”	98.6	3 ← AbsS
1	[same as 0]	97.1	4
2	[same as 0]	98.6	2
3	[same as 0, with quotation marks]	98.6	6
4	[same as 0]	98.6	2
5	“Two things are infinite: the universe and human stupidity; and I’m not sure about the universe.” – Albert Einstein	100.0	7 ← PCe, PSe
6	[same as 0]	97.1	2
7	[same as 0]	98.6	2

Table 17: AlpacaEval example 3 (Gemini-2.5-FL). **Prompt:** “What year was the Yamato Battleship built?” Judge scores on 0–7.

#	Response	AbsS	PS	Judge
0	The Yamato Battleship was built in 1941.	100	80.0	1 ← AbsS
1	The Yamato battleship was commissioned in 1941. Construction began in 1937.	95	90.0	3
2	The Yamato was laid down in 1937 and commissioned in 1941. Construction began in 1937; it entered service in 1941.	95	97.1	7 ← PSe
3	The Yamato battleship was laid down in 1937 and commissioned in 1941.	95	97.1	6
4	The Yamato battleship was commissioned in 1941.	95	79.3	2
5	The Yamato Battleship was built in 1941.	100	85.0	0
6	The Yamato Battleship was laid down in 1937 and commissioned in 1941.	95	97.1	4
7	The Yamato battleship was laid down in 1937 and commissioned in 1941.	95	97.1	5 ← PCe

proves Walker/GPT-4o-mini from 79.1 to 83.1 AUC ($p = .026$, correlated DeLong test) and Leenaars/Gemini-FL from 76.1 to 78.8 ($p = .004$), but its small Chausson gains are not significant ($p = .606$ and $.577$). These domains also differ in prevalence, prompt length, and criterion complexity, so the results support transfer across applications but do not identify discipline itself as a causal factor.

2. **Research-proposal ranking.** We use the expert judgments released by Si et al. (2024): 93 (48 human and 45 AI) generated NLP proposals across seven topics, each assigned an overall-quality score by expert reviewers. We sample 656 within-topic pairs and evaluate Claude Sonnet 4.5 against the mean expert score. In the cheap, single-order, non-CoT regime, for pairs separated by at least one expert-score point ($n = 427$), PairScore attains 60.7% accuracy

Table 18: AUC-ROC (%) for all eight models across four claim-verification datasets (same data as Table 2, grouped by dataset). **Bold** = best per row; * = significantly better than AbsScore(5x) at 95% (bootstrap).

Dataset	Model	AbsScore (1x)	AbsScore (5x)	PairChoice	PairScore
CREAK	GPT-5-mini	94.7	97.9	97.6	96.4
	GPT-4o-mini	95.7	96.5	96.1	97.5*
	GPT-4.1-nano	94.6	95.1	94.9	95.8
	Gemini-2.5-FL	95.0	96.7	93.8	97.1
	Qwen3-4B	88.9	89.7	83.2	91.9*
	Qwen2.5-3B	87.7	88.5	76.3	90.1
	Phi-4-mini	83.9	89.2	89.9	88.1
	Qwen3-1.7B	71.8	78.4	59.0	83.4*
FEVER	GPT-5-mini	94.9	95.6	94.3	93.7
	GPT-4o-mini	89.8	91.0	90.1	92.3
	GPT-4.1-nano	85.6	87.9	89.8*	90.2*
	Gemini-2.5-FL	90.1	91.4	85.9	91.1
	Qwen3-4B	81.3	81.7	77.7	86.1*
	Qwen2.5-3B	79.0	80.4	66.6	73.7
	Phi-4-mini	79.9	82.1	85.9*	82.3
	Qwen3-1.7B	62.3	63.1	49.4	71.7*
Dataset	Model	AbsScore (1x)	AbsScore (5x)	PairChoice	PairScore
TruthfulQA	GPT-5-mini	86.1	91.3	94.5*	92.0
	GPT-4o-mini	87.0	87.9	85.7	87.1
	GPT-4.1-nano	76.3	76.1	74.3	75.6
	Gemini-2.5-FL	86.5	87.1	88.5	91.7*
	Qwen3-4B	73.4	73.1	71.7	79.1*
	Qwen2.5-3B	65.3	65.2	72.4*	82.4*
	Phi-4-mini	63.8	65.1	67.1	74.7*
	Qwen3-1.7B	56.2	58.6	57.2	69.1*
CliniFact	GPT-5-mini	64.0	65.0	69.6*	71.1*
	GPT-4o-mini	60.1	62.1	58.2	63.4
	GPT-4.1-nano	52.7	53.8	59.5*	56.5
	Gemini-2.5-FL	60.7	60.4	59.4	54.4
	Qwen3-4B	56.4	56.4	50.4	56.5
	Qwen2.5-3B	55.1	57.8	61.1	50.5
	Phi-4-mini	52.3	54.6	52.0	55.7
	Qwen3-1.7B	48.8	47.0	48.4	55.8*

versus 50.6% for terse PairChoice; their win-rate rankings have within-topic Spearman correlations of .213 and .030, respectively. Note that PairChoice selects the second-presented proposal in all 656 calls, so randomized order turns this positional rule into chance-level accuracy.

In the CoT + order-counterbalanced control, we give both methods 2–4 sentences of reasoning space and explicitly query both A/B orders. PairChoice yields an order-consistent preference for 533/656 pairs (81.2%). On the 334 matched, non-tied pairs separated by at least one expert-score point, PairScore and PairChoice reach 58.7% and 57.5% accuracy, respectively (difference +1.2 points; 95% bootstrap CI $[-1.5, +3.9]$; McNemar $p = .503$). This control rules out a general accuracy advantage under equal scaffolding. It instead indicates that numerical pairwise scores are more robust in the terse, low-token regime, whereas binary comparison requires additional scaffolding and remains sensitive to order.

E Extended Related Work

Measurement theory and comparative judgment. Our work connects to the classical distinction between absolute and comparative judgment. [Thurstone \(1927\)](#) showed that reliable measurement scales can be constructed from pairwise

comparisons even when absolute ratings are unreliable. In psychophysics, [Stewart et al. \(2005\)](#) showed that even supposedly “absolute” identifications are driven by relative comparisons to recent stimuli, suggesting that comparative judgment is more fundamental than absolute rating. These findings motivate our hypothesis: if LLM confidence is inherently ordinal and context-dependent, pairwise elicitation is a more appropriate measurement tool than absolute numerical scoring.

Active ranking and noisy comparisons. [Jamieson and Nowak \(2011\)](#) show that adaptively chosen comparisons can identify rankings using $\tilde{O}(d \log n)$ queries in low-dimensional settings. [Maystre and Grossglauser \(2015\)](#) analyze sorting under the Bradley–Terry model; complementary results include noisy sorting ([Braverman and Mossel, 2008](#)) and minimax-optimal rates for noisy comparisons ([Shah and Wainwright, 2017](#)). The dueling-bandits framework ([Yue et al., 2009](#); [Bengs et al., 2021](#)) emphasizes that comparative feedback can be more reliable than absolute rewards.

Preference learning and RLHF. The success of Reinforcement Learning from Human Feedback (RLHF) ([Christiano et al., 2017](#); [Ouyang et al., 2022](#)) demonstrates that pairwise preferences are a reliable supervision signal. [Rafailov et al. \(2023\)](#) derive an implicit reward model from preferences

via Direct Preference Optimization (DPO), and [Azar et al. \(2024\)](#) clarify the relationship between preferences and rewards.

Inference-time compute scaling. Best-of- N sampling and related inference-time strategies have gained attention ([Snell et al., 2024](#)). Both numerical and pairwise confidence estimates can serve as selection criteria; our work compares these approaches.